

YULIIA FILIPPOVA

+32476320621

yuliiafilippova00@gmail.com

linkedin.com/in/yulii-filippova-9202a2279

SUMMARY

Data and AI specialist with hands-on experience gained through an NLP assistant role and ML/AI research internships. I hold an MA in Digital Text Analysis and am currently completing a postgraduate program in Applied AI. Now I'm looking for an early-career role where I can apply and further develop my skills. Eligible to work in Belgium (residence permit holder).

SKILLS

- Programming languages: Python (Pandas, NumPy, scikit-learn, spaCy, NLTK, HuggingFace, Matplotlib, TensorFlow, PyTorch), R, SQL (basic queries, joins);
- NLP and ML: Data preprocessing and analysis, model training and evaluation, text classification, sentiment analysis, embeddings, natural language identification, dataset annotation;
- Tools: LLMs and VLMs (Ollama, LM Studio, HuggingFace), MakeSense.ai, Flask, FastAPI, Git;
- Computer Vision: OpenCV, YOLO, Mediapipe;
- Languages: English (C1), Ukrainian (C2), Russian (C2), Dutch (B1);
- Soft Skills: Fast learner, adaptable, communicative, highly motivated, organized.

EDUCATION

ERASMUSHOGESCHOOL BRUSSEL
Applied Artificial Intelligence, PGDip
(2025-2026)

UNIVERSITY OF ANTWERP
Digital Text Analysis, MA
(2022-2024)

V.N. KARAZIN KHARKIV NATIONAL
UNIVERSITY
Languages, Literature and
Translation (English, German,
Ukrainian, Russian), BA (2017-2021)

WORK EXPERIENCE

AI Research Intern

Research Centre for Preventive Health Innovation (EhB)

March - present, 2026 | Belgium

- Developing an AI system for automated, time-aligned audio description generation to improve video accessibility for visually impaired users.

NLP/Data Assistant

Textgain

March - September, 2024 | Belgium

- Developed and evaluated ML approaches for a Native Language Identification project;
- Supported CaLICO LLM development for harmful content detection by collecting, processing and annotating social media data.

ML Research Intern

University of Antwerp (The Antwerp Centre for Digital Humanities and Literary Criticism (ACDC))

March - June, 2023 | Belgium

- Worked on the Silent Voices project processing medieval Dutch manuscripts encoded in TEI-XML;
- Extracted text from XML files and built a structured database;
- Using ML/NLP techniques developed an approach for expanding medieval Dutch abbreviations in collaboration with the research team.

PROJECTS

1. AI Audio Description Generation

- Developed an end-to-end pipeline to automatically generate audio descriptions for short-form videos;
- Combined scene detection, keyframe extraction and VLM+LLM for context-aware subtitle generation;
- Focused on accessibility by producing concise, time-aligned descriptions for visually impaired users;
- **GitHub:** <https://github.com/YuliiaFilippova/AI-AD-Generator>

2. Native Language Identification

- Developed ML approaches to identify the likely native language background of English-language writers using linguistic and syntactic n-gram features;
- Trained on multilingual Reddit data and evaluated on external English-language news corpora
- Best performance achieved with an SVM using word 1-grams and character 4-grams;

3. AI-Generated Text Detection

- Built a classifier using feature-based and transformer-based approaches to detect whether text was human-written or AI-generated;
- The dataset consists of human-written articles and texts generated via the OpenAI API;
- Achieved 97.3% accuracy with a fine-tuned BERT model.

4. Financial Sentiment Analysis

- Developed an end-to-end pipeline for sentiment classification of financial reports and news;
- It included text preprocessing, normalization, vectorization (TF-IDF), feature engineering, model selection (classical algorithms and LSTM) and evaluation;
- Reached 83% accuracy with an SVM classifier.

5. Computational Analysis of David Bowie's Music and Lyrics

- Conducted corpus-based linguistic and stylistic analysis of Bowie's discography across different artistic periods and alter egos, applying ML and NLP techniques;
- Collected lyrics via LyricsGenius API and metadata via Spotify API;

6. Master's Thesis: "Attributing Translatorship: A Computational Study of Translator Style of Constance Garnett and Charles Hogarth"

- Built feature-based ML classifiers to attribute translator identity, using extracted lexical, syntactic and stylistic features.

ADDITIONAL INFORMATION

- **Publication:** Co-authored an article with Professor Lukianova at V.N. Karazin Kharkiv National University (2022): <https://filol.dspu.in.ua/index.php/filol/article/view/215>
- Currently learning Dutch (B1 level);
- **Interests:** Accessibility subtitling and movie translation (English, Ukrainian, Russian), music, nature, cycling, travelling.